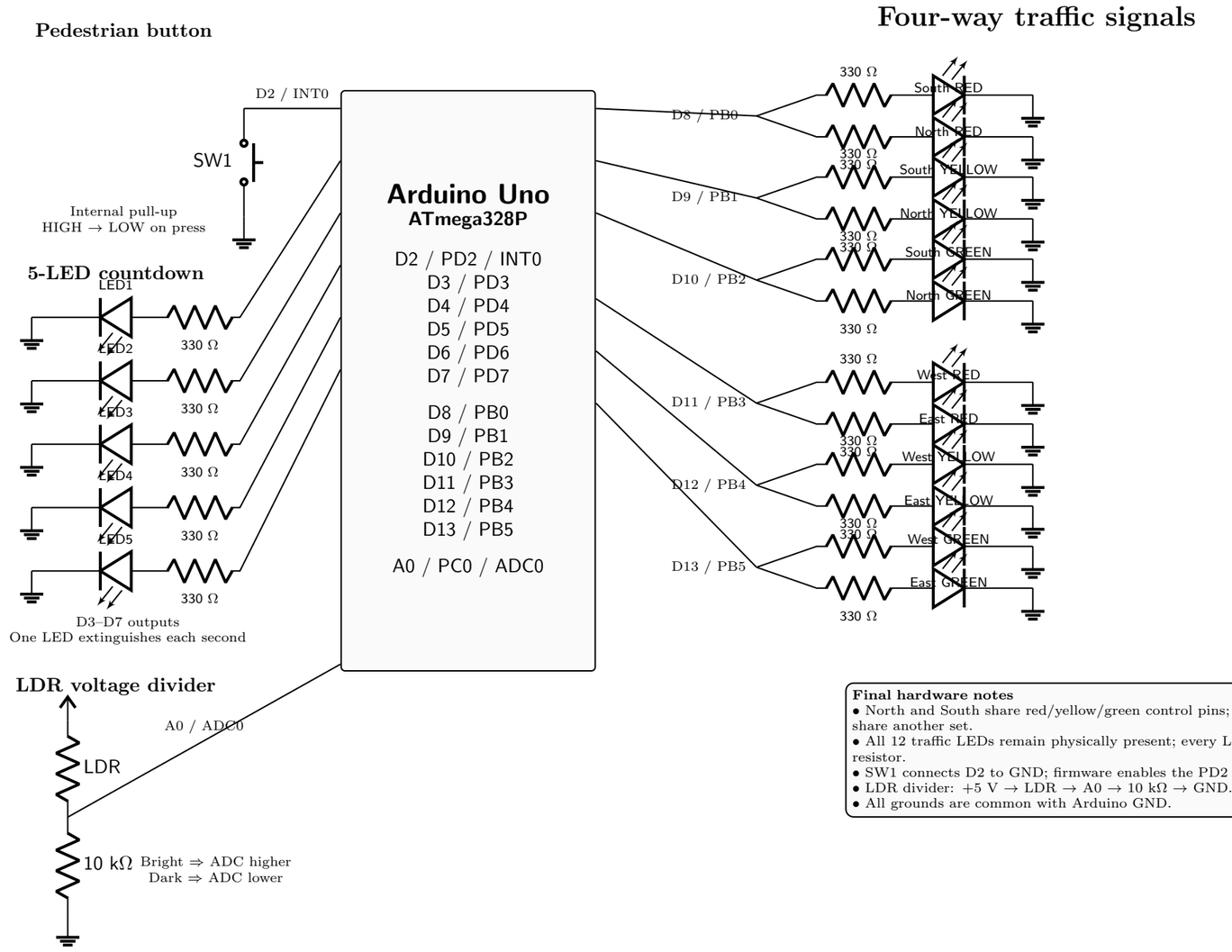
Smart Traffic Junction Controller — Complete Schematic

Arduino Uno / ATmega328P • Final 5-LED pedestrian countdown version



Final hardware notes

- North and South share red/yellow/green control pins; East and West share another set.
- All 12 traffic LEDs remain physically present; every LED has its own resistor.
- SW1 connects D2 to GND; firmware enables the PD2 internal pull-up.
- LDR divider: +5 V → LDR → A0 → 10 kΩ → GND.
- All grounds are common with Arduino GND.